

Anthony Katona

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🔄 Summary Statement

Chemistry PhD specializing in polymer and ceramic coating processes, photothermal materials processing, and materials characterization. Experienced in lab-scale process development, class IV laser systems, safety oversight, and custom software & electronics development towards productivity enhancement. Skilled at communication to a wide range of knowledge levels and building processes that improve transparency, productivity, and maintainability.

🎓 Experience & Education

The Pennsylvania State University

Chemistry (State College, PA)

2023-current

Postdoctoral Researcher

- 🔧 led laser-processing development for UHTC preceramic polymer coating precursors within an 8-group MURI collaboration, establishing experimental procedures, air-free workflows, and process-parameter screening methods
- 🔧 built remote monitoring instrumentation and a transparent flow/purge chamber for air-sensitive sample processing, improving real-time observation and control during laser treatment
- 🔧 designed nanosecond pyrometer; to inform collaborating theory groups of accessible peak temperatures during laser pulses
- 🔧 integrated containerization into multiple existing project software pipelines with complex dependency stacks to facilitate access to tools between collaborators

The Pennsylvania State University

Chemistry (State College, PA)

2023

PhD

- 🔧 developed photo-thermal polymer techniques for coatings & 3D-printing applications
- 🔧 modeled and printed custom 3DP components for managing optics and improving in-house HFR videography and contact angle goniometry
- 🔧 synthesized ultra-fast/nano-scale thermal transport materials
- 🔧 developed Python analysis pipelines for profilometry, mechanical testing, DMA, and thermal datasets, converting raw instrument outputs into reproducible plots, metrics, and comparison-ready summaries
- ⚙️ oversaw lab safety as both General & Laser Safety Officer, including management of chemical inventory and coordinating laboratory audits
- ⚙️ managed supply & maintenance of pumps, gas cylinders, and gloveboxes
- 👤 taught as TA for all 5 available first year chemistry courses for a total of 8 semesters
- 👤 co-developed & implemented novel thermal conductivity undergraduate lab experiments for *Dr. Bratoljub Milosavljevic's* physical chemistry course
- 👤 served as graduate student TA mentor for 1/3 of class & research mentor for 5 undergraduate researchers, teaching experimental design, instrument use, scripting for data analysis, and scientific writing

Susquehanna University

Biochemistry; Physics minor (Selinsgrove, PA)

2015

BS

- 🔧 synthesized transition metal Janus complexes with *Dr. William Dougherty* (thesis project)
- 🔧 assisted in galaxy classification model training *Dr. Violet Mager* (summer)

Technical Skills

 **spectroscopy:** NMR, IR, UV-Vis

 **microscopy:** AFM, TEM (STEM, EDS)

 **materials:** mechanical (UTM, DMA) & thermal (DSC, TGA, TPS) tests; lathes, mills, 3DP, engravers

 **laser:** class IV CW/pulsed safety, handling, & optics

 **electronics:** circuit design/prototyping, signal analysis, micro-controllers

 **languages:** Python, C#, C++, LabVIEW, Blueprints, HTML, CSS, JS, SQL, LaTeX, markdown

 **data and automation:** Python, Pandas/NumPy/SciPy, ImageJ, WEKA segmentation, Veusz, LabVIEW, SQL, Docker, Arduino/micro-controllers  **technical communication:** Adobe CC (Ae, Il, Ps, Pt, Pr), FL Studio, Blender 3D, Unreal Engine

External Resources

publications:

- S Phillips *et al*, 2026 (review). Indust. Chem. & Mat, **“Nanometer-Scale Surface Roughness...”**
- A Katona & B Lear, 2025. J. Phys. Chem. C , **“Photothermal Curing of Polydimethylsiloxane: Carbon Black Composites Results in Changes to Polymer Topography, Cross-Link Density, and Mass Density”**
- A Blasone *et al*, 2025. **“Rapid Photothermal Curing of PDMS on Paper”**
- A Katona & B Lear, 2024. Macromol., **“Effective photothermal curing of PDMS using ultra-low loadings of carbon black ”**
- V Mager *et al*, 2018, ApJ 864 **“Galaxy Structure in the Ultraviolet...”**

patent filings:

- Co-inventor, “Photothermal Generation of Ceramics,” U.S. Provisional Patent Application No. 63/910,435, filed Nov. 3, 2025.
- Co-inventor, “Photothermal Curing of Polymers on Thermally Sensitive Substrates,” U.S. patent application pending; priority applications filed July, November, and December 2024.

portfolio: frkatona.github.io

references:

- Dr. Benjamin Lear, Professor of Chemistry (advisor) - bul14@psu.edu
- Dr. Mary Jo Bojan, Coordinator of General Chemistry (supervisor) - bz6@psu.edu
- Dr. Zackary Rhoden, Professor of Chemistry (colleague) - zcr5008@psu.edu